

Mohammad Ratul Mahjabin

mohammadratul@usf.edu | www.ratul-mahjabin.github.io

Education

University of South Florida, Tampa, FL

Ph.D. in Computer Science and Engineering

Jan '24 - Current

Research Areas: Computational Social Science, Human-Computer Interaction, NLP

Advisor: Dr. Raiyan Abdul Baten - [Computational Social Science and AI Lab](#)

Islamic University of Technology, Dhaka, Bangladesh

B.Sc. in Software Engineering

Jan '19 - May '23

Thesis: [An Efficient Deep Learning-based approach for Recognizing Agricultural Pests in the Wild](#)

Advisor: Assist. Prof. Sabbir Ahmed

Publications

Recent preprints under review

- P1 **M. R. Mahjabin** and R. A. Baten, [The Wisdom of Intellectually Humble Networks](#), [arxiv:2502.02015](#), 2025

Journal Papers

- J2 Rafi TH, **Mahjabin R**, Ghosh E, Ko YW, Lee JG. [Domain generalization for semantic segmentation: a survey](#). Artificial Intelligence Review. 2024 Aug 12;57(9):247.
- J1 Islam MN, Jahangir R, Mohim NS, Wasif-UI-Islam M, Ashraf A, Khan NI, **Mahjabin MR**, Miah AS, Shin J. [A multilingual handwriting learning system for visually impaired people](#). IEEE Access. 2024 Jan 15.

Conference Papers

- C2 Hema NE, Zeest NS, **Mahjabin MR**, Islam MN. [An Ensemble-based ML Model to Predict Cardiac Disease](#). In 2024 International Conference on Inventive Computation Technologies (ICICT) 2024 Apr 24 (pp. 214-219). IEEE
- C1 Rafi MH, **Mahjabin MR**, Rahman MS, Kabir MH, Ahmed S. [A Critical Analysis of Deep Learning Applications in Crop Pest Classification: Promising Pathways and Limitations](#). In 2023 26th International Conference on Computer and Information Technology (ICCIT) 2023 Dec 13 (pp. 1-6). IEEE.

Professional Experience

University of South Florida, Tampa, FL

Graduate Research and Teaching Assistant, Dept. of Computer Science and Engineering

Jan '24 - Current

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

Research Assistant, Dept. of Computer Science and Engineering

Aug '23 - Dec '23

Samsung R&D Institute, Dhaka, Bangladesh

Software Engineer Intern

May '22 - Sept '22

Teaching

University of South Florida

Teaching Assistant

- CAI5133: Social Media Mining
- COP4530: Data Structures
- CIS4250: Ethical Issues/ Profess Conduct

Spring '25
Summer '24
Spring, Fall '24

Advising

Undergraduate Research Mentees

- Nur E Jannat Hema (Computer Science and Engineering, MIST, 2023)
- Nubras Samayeen Zeest (Computer Science and Engineering, MIST, 2023)

Selected Projects

PPD Coach

Mobile Application

An android application for Screening Postpartum Depression in the context of Bangladesh

2023

- It screens postpartum depression using story-based questions and visual aids.
- Developed with: [Flutter](#), [Firebase](#)

Software Defect Prediction

Notebook

Software fault prediction using machine learning [\[Github\]](#)

2022

- It predicts defects in software modules using different ML models
- Developed with: [Python](#), [Pandas](#), [Numpy](#)

HaateKhor

Web Application

ML-powered web app for early childhood education [\[Github\]](#)

2021

- This project utilizes ML techniques to provide early education to children in an engaging and interactive way [\[Video Demo\]](#)
- Developed with: [Python](#), [OpenCV](#), [YOLO](#), [React](#), [Nodes.js](#)

Covid-CXRNet

Notebook

Detecting Covid19 with Chest X-ray using CNN [\[Github\]](#)

2021

- This project project implements a CNN model(ResNet18) to accurately detect COVID-19 with an accuracy of 99.98%
- Developed with: [Python](#), [Pytorch](#)

Interests

Computational Social Science, Social Media Mining, Machine Learning, Data Science, Network Analysis, Data Mining, Human-Computer Interaction, Natural Language Processing, Statistical Modeling

Skills

Programming	Python, R, C/C++, Javascript, Java, HTML, CSS
Libraries	PyTorch, TensorFlow, Keras, OpenCV, NLTK, Pandas, Scikit-Learn, Matplotlib, Numpy
Web & DB	Flask, Django, React, MongoDB, Firebase, REST API
Tool	Docker, Heroku, Git, LaTeX